

Cycles refute infinite log-concavity for chromatic coefficients

Exact thresholds, finite certificates, and an exponential boundary limit

Research report prepared for Vladimir Reshetnikov
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Abstract

Amdeberhan and Moll's Conjecture 21 asserts that the absolute coefficients of every chromatic polynomial form an infinitely log-concave sequence. We give an elementary disproof using cycle graphs, and classify the entire cycle family. With zero padding and $\mathcal{L}(a)_k = a_k^2 - a_{k-1}a_{k+1}$, the absolute coefficient sequence of C_n is infinitely log-concave exactly when $3 \leq n \leq 11$. Its first negative iterate occurs at depth 5 for $n = 12$, depth 4 for $13 \leq n \leq 16$, and depth 3 for all $n \geq 17$. Every cycle sequence is nevertheless 2-log-concave. The infinite negative family follows from a factored degree-16 polynomial; the positive cases are certified by an invariant 3-factor log-concavity region. A normalized limit with coefficients of $e^x - 1$ explains the failure and gives a perturbation-stable obstruction. We also derive generating functions and a constant-coefficient recurrence for the integer witness sequence. All finite checks use exact integers or rational numbers. The report does not claim priority over all existing literature, nor minimality among graphs other than cycles.

A compact disproof. For C_{17} , the third iterate has

$$(\mathcal{L}^3 a^{(n)})_2 = 22\,491\,136^2 - 65\,536 \cdot 8\,150\,077\,440 = -28\,272\,276\,537\,344 < 0.$$

The complete local calculation appears in Section 3; no floating-point calculation or exhaustive graph search is required.

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1 The problem, conventions, and main result

1.1 A precise published conjecture

The target is Conjecture 21 in *Infinite log-concavity* by Amdeberhan and Moll [1]: the absolute values of the coefficients of any chromatic polynomial should be infinitely log-concave. The statement appears on printed page 9 of the publisher's PDF; bibliographic and manuscript-date details are recorded in the references.

This is a stronger assertion than ordinary log-concavity of chromatic coefficients, established by Huh [4]. A different sufficient condition comes from Brändén [3]: a polynomial with nonnegative coefficients and only real nonpositive zeros has an infinitely log-concave coefficient sequence. Neither theorem implies the proposed statement for all chromatic polynomials.

Status and attribution. The primary paper inspected on September 19, 2026 still formulates the assertion as a conjecture. A targeted search did not locate a prior cycle counterexample, but it was not an exhaustive bibliographic search. Thus this report establishes a disproof of the stated formulation, without claiming to certify that it remained unresolved everywhere until this report. The invariant-cone method used below is established machinery, attributed to McNamara and Sagan [2]; its proof is included.

1.2 The nonlinear operator

Definition 1.1. For a finite sequence $a = (a_0, \dots, a_N)$, set $a_k = 0$ when $k < 0$ or $k > N$, and define

$$(\mathcal{L}a)_k = a_k^2 - a_{k-1}a_{k+1}. \quad (1)$$

A nonnegative sequence is *r-log-concave* when $\mathcal{L}^j a$ is entrywise nonnegative for every $1 \leq j \leq r$. It is *infinitely log-concave* when this holds for every $j \geq 1$.

Zero extension is the convention explicitly used in [2]. It keeps both endpoints: the first and last nonzero entries are squared, rather than discarded, at each iteration. In particular, we do not take absolute values again after applying $\mathcal{L}$.

For a sequence under consideration, define its *first failure depth* by

$$h(a) = \min\{j \geq 1 : \mathcal{L}^j a \text{ has a negative entry}\}, \quad (2)$$

with $h(a) = \infty$ if this set is empty. Write $h(n) = h(a^{(n)})$ for the absolute coefficient sequence of the simple cycle C_n , indexed by increasing degree.

Theorem 1.2 (Complete cycle classification). *For every integer $n \geq 3$, the absolute coefficients of the chromatic polynomial of C_n form the sequence*

$$a_0^{(n)} = 0, \quad a_1^{(n)} = n - 1, \quad a_k^{(n)} = \binom{n}{k} \quad (2 \leq k \leq n). \quad (3)$$

Every such sequence is 2-log-concave, and

$$h(n) = \begin{cases} \infty, & 3 \leq n \leq 11, \\ 5, & n = 12, \\ 4, & 13 \leq n \leq 16, \\ 3, & n \geq 17. \end{cases} \quad (4)$$

In every finite case, the coefficient of degree 2 is negative at the first failure depth. In particular, C_{12} is the smallest cycle counterexample to the infinite-log-concavity conjecture.

The qualification “cycle” in the last sentence matters. No exhaustive search through all graphs on at most eleven vertices is asserted here. The proof combines a symbolic argument valid for every $n \geq 17$ with a small collection of exact certificates for $3 \leq n \leq 16$.

2 The cycle polynomial and its coefficient triangle

Proposition 2.1. *For $n \geq 3$, the chromatic polynomial and its absolute-coefficient polynomial are respectively*

$$P_{C_n}(q) = (q-1)^n + (-1)^n(q-1), \quad (5)$$

$$A_n(x) = \sum_{k=0}^n a_k^{(n)} x^k = (1+x)^n - (1+x). \quad (6)$$

Proof. The triangle has $q(q-1)(q-2)$ proper colorings, which agrees with (5). For $n \geq 4$, delete one edge of C_n . The remaining path has $q(q-1)^{n-1}$ proper colorings. The colorings in which the two endpoints receive the same color are precisely those of the contracted cycle C_{n-1} . Therefore

$$P_{C_n}(q) = q(q-1)^{n-1} - P_{C_{n-1}}(q).$$

Substituting the induction hypothesis proves (5). The constant coefficient cancels, the linear coefficient has magnitude $n-1$, and all coefficients of degree at least two are alternating binomial coefficients. Equivalently, $(-1)^n P_{C_n}(-x) = (1+x)^n - (1+x)$ has nonnegative coefficients, proving (6) and (3). $\square$

Thus the sequences are almost rows of Pascal’s triangle: their constant coefficient is replaced by zero and their linear coefficient is reduced by one. An arbitrarily small relative change in the latter coefficient does not make the nonlinear iteration harmless, because the change at the constant coefficient is not small.

The row polynomials satisfy two useful recurrences:

$$A_{n+1}(x) = (1+x)A_n(x) + x(1+x), \quad (7)$$

$$A_n(x) = A_{n-1}(x) + x(1+x)^{n-1} \quad (n \geq 4). \quad (8)$$

The second is also a convenient independent way to generate the coefficient triangle. At the coefficient level, (7) is Pascal’s recurrence with a correction of 1 in degrees 1 and 2:

$$a_k^{(n+1)} = a_k^{(n)} + a_{k-1}^{(n)} + \mathbf{1}_{k=1} + \mathbf{1}_{k=2}.$$

The bivariate ordinary generating function is rational:

$$\sum_{n \geq 3} A_n(x) t^n = \frac{(1+x)^3 t^3}{1 - (1+x)t} - \frac{(1+x)t^3}{1-t}. \quad (9)$$

These identities also specify the integer triangle in the bundled data without relying on an external sequence identifier.

Indexing is not the obstruction. For a shift T_s on zero-extended sequences, $(T_s a)_k = a_{k-s}$, we have $\mathcal{L}T_s = T_s \mathcal{L}$. The operator also commutes with reversal of a finite coefficient list. Consequently the counterexample persists when coefficients are listed in decreasing degree or when the initial zero coefficient is omitted and the remaining entries are reindexed. Deleting endpoints at *every* iteration, by contrast, would define a different operator.

3 A short exact counterexample: the seventeen-cycle

Only six input coefficients are needed to compute $(\mathcal{L}^3 a)_2$. For C_{17} they are

$$(a_0, a_1, a_2, a_3, a_4, a_5) = (0, 16, 136, 680, 2380, 6188).$$

Let $b = \mathcal{L}a$, $c = \mathcal{L}^2 a$, and $d = \mathcal{L}^3 a$. Table 1 contains every nontrivial intermediate quantity needed for d_2 . Blank positions are unnecessary, not assumed to be zero.

Iterate	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
a	0	16	136	680	2380	6188
b	0	256	7616	138720	1456560	
c	0	65536	22491136	8150077440		

Table 1: An exact dependency triangle for the counterexample C_{17} .

For example,

$$\begin{aligned} b_2 &= 136^2 - 16 \cdot 680 = 7616, \\ b_3 &= 680^2 - 136 \cdot 2380 = 138720, \\ b_4 &= 2380^2 - 680 \cdot 6188 = 1456560, \\ c_2 &= 7616^2 - 256 \cdot 138720 = 22491136, \\ c_3 &= 138720^2 - 7616 \cdot 1456560 = 8150077440. \end{aligned}$$

Since $a_0 = b_0 = c_0 = 0$, the first positive entry is repeatedly squared, so $c_1 = 16^4 = 65536$. Finally,

$$\begin{aligned} d_2 &= 22491136^2 - 65536 \cdot 8150077440 \\ &= 505851198570496 - 534123475107840 \\ &= -28272276537344 < 0. \end{aligned} \tag{10}$$

This exact calculation by itself disproves the universal conjecture. The remaining sections explain why the example works, prove an infinite family, and identify the smaller cycles that fail only at later depths.

4 Symbolic iterates and the threshold at seventeen

Throughout this section, $a = a^{(n)}$, $b = \mathcal{L}a$, $c = \mathcal{L}^2 a$, and $d = \mathcal{L}^3 a$. We compare them with a full Pascal row $p_k = \binom{n}{k}$ and its successive iterates $B = \mathcal{L}p$, $C = \mathcal{L}^2 p$, and $D = \mathcal{L}^3 p$.

4.1 The first two Pascal iterates

For $0 \leq k \leq n$, direct cancellation of binomial ratios gives

$$B_k = p_k^2 \frac{n+1}{(k+1)(n-k+1)}, \tag{11}$$

$$C_k = B_k^2 \frac{2(n+2)}{(k+2)(n-k+2)}. \tag{12}$$

For interior indices, the corresponding neighboring-product ratios are

$$\frac{B_{k-1}B_{k+1}}{B_k^2} = \frac{k(n-k)}{(k+2)(n-k+2)}, \quad (13)$$

$$\frac{C_{k-1}C_{k+1}}{C_k^2} = \frac{k^2(n-k)^2}{(k+1)(k+3)(n-k+1)(n-k+3)}. \quad (14)$$

Both ratios are strictly less than one. At either endpoint the neighboring product is zero, and the same positivity conclusion holds. In particular $B_k, C_k, D_k > 0$ for $0 \leq k \leq n$. These elementary identities suffice for the tail comparisons below; no general real-rootedness theorem is needed in the proof of the cycle classification.

4.2 The exceptional coefficients

Because $a_0 = 0$, we have $b_0 = c_0 = d_0 = 0$. The first iterate begins with

$$b_1 = (n-1)^2, \quad (15)$$

$$b_2 = \frac{n(n-1)^2(n+4)}{12}, \quad (16)$$

$$b_k = B_k \quad (3 \leq k \leq n). \quad (17)$$

For later calculations the two specific tail values are

$$b_3 = \frac{n^2(n-1)^2(n-2)(n+1)}{144}, \quad (18)$$

$$b_4 = \frac{n^2(n-1)^2(n-2)^2(n-3)(n+1)}{2880}. \quad (19)$$

The formula for b_4 is also valid at $n = 3$, where it is zero by padding. All entries on the positive support of b are positive.

The second iterate has

$$c_1 = (n-1)^4, \quad (20)$$

$$c_2 = \frac{n^2(n-1)^4(n+2)}{16}, \quad (21)$$

$$c_3 = \frac{n^3(n-2)^2(n-1)^4(n+1)(n^2+n+18)}{51840}, \quad (22)$$

$$c_k = C_k \quad (4 \leq k \leq n). \quad (23)$$

For instance, $c_2 = b_2^2 - b_1b_3$ reduces to

$$\frac{n^2(n-1)^4}{144}((n+4)^2 - (n-2)(n+1)) = \frac{n^2(n-1)^4(n+2)}{16}.$$

Substitution of (16)–(19) gives (22). The exact algebra is also audited by `derive_symbolic.py`.

Proposition 4.1. *For every $n \geq 3$, the cycle coefficient sequence is 2-log-concave. In fact, both its first and second iterates are strictly positive on indices $1, \dots, n$.*

Proof. Every factor in (15)–(17) and (20)–(23) is positive on the indicated support. The remaining entries are zero. The positivity of the Pascal terms was proved using (11) and (12). $\square$

4.3 The decisive quartic

Define

$$Q(n) = 2n^4 - 12n^3 - 327n^2 - 412n + 36. \quad (24)$$

Using $d_2 = c_2^2 - c_1c_3$, we obtain

$$d_2 = -\frac{n^3(n-1)^8(n+4)}{103680} Q(n). \quad (25)$$

An explicit intermediate form, useful for checking the factorization, is

$$d_2 = \frac{n^3(n-1)^8}{103680} (405n(n+2)^2 - 2(n-2)^2(n+1)(n^2+n+18)).$$

The polynomial in parentheses is exactly $-(n+4)Q(n)$.

Lemma 4.2. *For integral $n \geq 3$, $Q(n) < 0$ when $n \leq 16$, and $Q(n) > 0$ when $n \geq 17$.*

Proof. On $3 \leq n \leq 16$, the increasing quadratic $2n^2 - 12n - 327$ is at most its value -7 at $n = 16$. Hence

$$Q(n) = n^2(2n^2 - 12n - 327) - 412n + 36 < 0.$$

For $n = 17 + m$, $m \geq 0$, expansion gives

$$Q(17 + m) = 2m^4 + 124m^3 + 2529m^2 + 17370m + 6615 > 0. \quad (26)$$

□

Corollary 4.3. *For every $n \geq 17$, $a^{(n)}$ fails at its third iterate. In particular, the conjecture has infinitely many connected, simple counterexamples.*

Proof. The prefactor outside $Q(n)$ in (25) is negative and nonzero. Lemma 4.2 therefore gives $d_2 < 0$. Proposition 4.1 shows that no earlier failure is possible. □

4.4 No other third-iterate obstruction for smaller cycles

To prove exact failure depths for $12 \leq n \leq 16$, it remains to know that their *entire* third iterates are nonnegative, not just d_2 . Define

$$S(n) = 4n^5 - 27n^4 + 186n^3 + 145n^2 + 684n - 3564.$$

Direct calculation gives

$$d_3 = \frac{n^6(n-2)^3(n-1)^8(n+1)^2}{10749542400} S(n). \quad (27)$$

Every $n \geq 3$ has $S(n) > 0$, since

$$S(3 + m) = 4m^5 + 33m^4 + 222m^3 + 1441m^2 + 5280m + 3600. \quad (28)$$

The next coordinate admits a useful comparison rather than a large factorization. From the first-iterate formulas,

$$b_2 = B_2 + p_3, \quad c_3 = C_3 - p_3B_4 \leq C_3.$$

Since $c_k = C_k$ for $k \geq 4$, it follows that

$$d_4 = C_4^2 - c_3C_5 \geq C_4^2 - C_3C_5 = D_4 > 0 \quad (4 \leq n), \quad (29)$$

where $C_5 = 0$ when $n = 4$. For $k \geq 5$ on the support, $d_k = D_k > 0$. Finally $d_1 = (n-1)^8 > 0$.

Proposition 4.4. *The cycle coefficient sequence is 3-log-concave exactly when $3 \leq n \leq 16$. In that range its third iterate is strictly positive on the nonzero support.*

Proof. The formulas above establish positivity at every index other than 2; Lemma 4.2 and (25) settle index 2. For $n = 3$, indices beyond 3 are simply absent. Proposition 4.1 supplies the earlier two iterates. $\square$

5 Infinite positivity for the nine small cycles

A long list of positive iterates is not a proof of infinite log-concavity. Instead we use a region that $\mathcal{L}$ maps into itself. The following sufficient criterion is the McNamara–Sagan invariant-cone argument [2, Lemma 2.1]; using the rational constant 3 makes the certificates especially simple.

Lemma 5.1 (Invariant factor-log-concavity region). *Let $r \geq (3 + \sqrt{5})/2$ and let a be nonnegative with*

$$a_k^2 \geq r a_{k-1} a_{k+1} \quad \text{for all } k. \quad (30)$$

Then $b = \mathcal{L}a$ is nonnegative and satisfies the same inequalities. Consequently a is infinitely log-concave. In particular, $r = 3$ is sufficient.

Proof. The assumption gives, coordinatewise,

$$(1 - 1/r)a_k^2 \leq b_k \leq a_k^2, \quad b_k \geq 0.$$

Therefore

$$\begin{aligned} b_k^2 &\geq (1 - 1/r)^2 a_k^4 \\ &\geq (r - 1)^2 a_{k-1}^2 a_{k+1}^2 \\ &\geq r b_{k-1} b_{k+1}. \end{aligned}$$

The last inequality uses $(r - 1)^2 \geq r$ and $0 \leq b_{k\pm 1} \leq a_{k\pm 1}^2$. All steps remain valid when a coordinate is zero; no division by an entry of a is used. Iteration proves the claim. $\square$

For a nonnegative vector u with at least one positive neighboring product, let

$$R(u) = \min_{k: u_{k-1}u_{k+1} > 0} \frac{u_k^2}{u_{k-1}u_{k+1}}.$$

If $R(u) \geq 3$, indices with zero neighboring product satisfy the cone condition automatically. Table 2 gives the exact minimum for a suitable iterate of each small cycle.

The entries are obtained from (3) by applying (1) at most three times and comparing fractions by cross multiplication. The complete vectors are provided in `data/certificates.json`; Appendix B gives a short, independent-of-SymPy checker of all these minima. Thus each row represents finitely many explicit integer inequalities, not an extrapolation from a numerical plot.

Proposition 5.2. *The sequences $a^{(n)}$ are infinitely log-concave for $3 \leq n \leq 11$.*

Proof. All iterates preceding those in Table 2 are nonnegative by Propositions 4.1 and 4.4, or by the initial row. The tabulated iterate is nonnegative and satisfies the $r = 3$ condition. Lemma 5.1 certifies every subsequent iterate. $\square$

n	Iterate j	Minimizing index k	$R(\mathcal{L}^j a^{(n)})$
3	0	2	9/2
4	1	3	25/6
5	2	3	256/49
6	2	3	100/27
7	2	3	6845/2187
8	3	3	30926/5863
9	3	3	8736525/1666093
10	3	3	2991871581875/519958761651
11	3	2	32433025/8287488

Table 2: Finite certificates of infinite log-concavity. Every displayed ratio is greater than 3. All ratios and minima are exact.

It is worth distinguishing a certificate from a necessary condition: this proof does not assert that every infinitely log-concave sequence eventually enters this cone. Only the nine displayed cases require that property here, and for them it is checked exactly.

6 The delayed failures at twelve through sixteen

For $12 \leq n \leq 16$, Proposition 4.4 shows that depth three is still positive. The next calculations determine the exact first failure. In Table 3, the middle column is the first failure depth, and the final column is the offending entry of that iterate.

n	$h(n)$	$(\mathcal{L}^{h(n)} a^{(n)})_2$
12	5	-249621701601023742801969101519201265582387397744
13	4	-3618341131654935620812800
14	4	-199158562975246657489530096
15	4	-2734560032125157358883149375
16	4	-2598266861840295000000000000
17	3	-28272276537344

Table 3: Exact negative witnesses. The $n = 17$ row is repeated for comparison with the delayed failures.

For $n = 13, 14, 15, 16$, one can reproduce the values without constructing a full fourth-iterate vector: use

$$(\mathcal{L}^4 a)_2 = d_2^2 - d_1 d_3 \quad (31)$$

with $d_1 = (n-1)^8$ and the formulas (25), (27). Their earlier iterates are already known to be positive, so these negative values settle their exact depths.

For $n = 12$, all fourth-iterate entries are nonnegative. The complete fourth-iterate vector, with every integer explicitly displayed, appears in Appendix A. Its first three positive entries are

$$\begin{aligned} e_1 &= 45949729863572161, \\ e_2 &= 1505318939996586804452286, \\ e_3 &= 54746933663864342589859620161140. \end{aligned}$$

Consequently

$$(\mathcal{L}^5 a^{(12)})_2 = e_2^2 - e_1 e_3 = -249621701601023742801969101519201265582387397744 < 0.$$

This also explains why checking only three or four iterations would miss some counterexamples.

Completion of Theorem 1.2. Proposition 2.1 gives the coefficient row; Proposition 4.1 gives universal 2-log-concavity; Corollary 4.3 gives $h(n) = 3$ for $n \geq 17$; Proposition 5.2 gives the nine infinite cases; and the exact calculations in this section give all remaining depths. These cases exhaust $n \geq 3$. $\square$

7 Why the failure is structural: an exponential limit

7.1 Locality and geometric normalization

For $t > 0$, define the geometric rescaling $(G_t a)_k = t^k a_k$. Directly from (1),

$$\mathcal{L}^j(G_t a)_k = t^{2^j k} (\mathcal{L}^j a)_k. \quad (32)$$

This follows by induction, since each iteration squares the scale. Moreover, $(\mathcal{L}^j a)_k$ is a polynomial in the entries $a_{k-j}, \dots, a_{k+j}$, with negative-index entries fixed at zero. Each iteration enlarges the dependency interval by one on each side.

Taking $t = 1/n$ gives a normalized row $u_k^{(n)} = a_k^{(n)} / n^k$. For fixed k ,

$$u_0^{(n)} = 0, \quad u_1^{(n)} \longrightarrow 1, \quad u_k^{(n)} \longrightarrow \frac{1}{k!} \quad (k \geq 2).$$

Thus the coefficientwise limit is the sequence u of $e^x - 1$:

$$u = (0, 1, 1/2!, 1/3!, 1/4!, 1/5!, \dots). \quad (33)$$

The necessary local iterates of this limiting sequence are particularly simple:

$$\begin{aligned} u &= (0, 1, 1/2, 1/6, 1/24, 1/120, \dots), \\ \mathcal{L}u &= (0, 1, 1/12, 1/144, 1/2880, \dots), \\ \mathcal{L}^2 u &= (0, 1, 0, 1/51840, \dots). \end{aligned}$$

The zero between two positive entries forces

$$(\mathcal{L}^3 u)_2 = -\frac{1}{51840}. \quad (34)$$

Only a finite prefix is involved, so passage to the limit follows from continuity of a polynomial; there is no interchange of an infinite number of nonlinear iterations with a limit.

Theorem 7.1 (A local, perturbation-stable obstruction). *Let $v^{(N)}$ be nonnegative sequences indexed by the nonnegative integers, zero-extended to negative indices. Suppose $s_N > 0$ and*

$$v_0^{(N)} \longrightarrow 0, \quad \frac{v_k^{(N)}}{s_N^k} \longrightarrow \frac{1}{k!} \quad (1 \leq k \leq 5).$$

Then

$$s_N^{-16} (\mathcal{L}^3 v^{(N)})_2 \longrightarrow -\frac{1}{51840}. \quad (35)$$

In particular, all sufficiently late members of the family fail 3-log-concavity.

Proof. Apply (32) with $t = 1/s_N$ and $j = 3, k = 2$. The required entry depends only on indices -1 through 5 . Those entries converge to the corresponding ones in (33), with the negative index fixed at zero. Polynomial continuity and (34) prove the limit. Its strict negativity gives an eventually negative entry. $\square$

For example, for any fixed real λ , the rows

$$(0, n - \lambda, \binom{n}{2}, \dots, \binom{n}{n})$$

are nonnegative for all sufficiently large n and fail 3-log-concavity eventually. More generally, the same conclusion survives arbitrary perturbations $o(n^k)$ in coordinates $1 \leq k \leq 5$, provided the zeroth coordinate tends to zero and the resulting rows remain nonnegative. This is a statement about local coefficient data, not about whether every such perturbation is realized by a chromatic polynomial.

7.2 A quantitative asymptotic expansion

Equation (25) sharpens the limiting argument to an exact polynomial. Expanding its highest powers gives

$$(\mathcal{L}^3 a^{(n)})_2 = -\frac{n^{16}}{51840} + \frac{n^{15}}{5184} + \frac{287n^{14}}{103680} + O(n^{13}). \quad (36)$$

The remainder is simply the lower-degree part of a polynomial, not an uncontrolled approximation. The asymptotic limit explains eventual failure, whereas the positive-shift quartic (26) proves the exact threshold 17.

The mechanism is a boundary effect. The full Pascal row, after the same normalization, tends to the coefficients of e^x , whose constant term is 1. Our rows instead tend to $e^x - 1$. The changed boundary produces a zero in the second iterate and a strict negative entry in the third. It would therefore be unsafe to transfer infinite log-concavity from Pascal rows merely because most coefficients are unchanged.

8 An integer witness sequence, its recurrence and generating function

The negative obstruction itself defines a natural positive integer sequence. Set

$$w_m = -(\mathcal{L}^3 a^{(17+m)})_2 \quad (m \geq 0). \quad (37)$$

From (25) and (26),

$$w_m = \frac{(m+17)^3(m+16)^8(m+21)}{103680} \times (2m^4 + 124m^3 + 2529m^2 + 17370m + 6615). \quad (38)$$

Integrality follows directly from the integer sequence recurrence defining $\mathcal{L}$; it need not be inferred from the rational prefactor.

The first three values are

$$28272276537344, \quad 229969795557447, \quad 880220808750663.$$

Equation (38) expresses w_m as a degree-16 polynomial in m , with strictly positive coefficients and leading coefficient $1/51840$. In particular it is strictly increasing for $m \geq 0$.

Let $\Delta f(m) = f(m+1) - f(m)$ and let $\delta_j = \Delta^j w_0$. Then Newton's finite expansion gives

$$w_m = \sum_{j=0}^{16} \delta_j \binom{m}{j}. \quad (39)$$

Every δ_j for $0 \leq j \leq 16$ is a positive integer. To see positivity, express the polynomial in powers m^r : the coefficients are positive, and $\Delta^j m^r|_{m=0} = j! \left\{ \begin{smallmatrix} r \\ j \end{smallmatrix} \right\} \geq 0$. For $j = 0$ the positive constant term suffices; for $j \geq 1$ the m^j coefficient already contributes positively.

Consequently the ordinary generating function is

$$W(t) = \sum_{m \geq 0} w_m t^m = \sum_{j=0}^{16} \frac{\delta_j t^j}{(1-t)^{j+1}} = \frac{P(t)}{(1-t)^{17}}, \quad (40)$$

where an exact specification of the degree-at-most-16 numerator is

$$P(t) = \sum_{j=0}^{16} \delta_j t^j (1-t)^{16-j}, \quad \delta_j = \sum_{i=0}^j (-1)^{j-i} \binom{j}{i} w_i. \quad (41)$$

These are finite closed formulas, rather than auxiliary recurrences needed to define the coefficients. The integer lists δ_j and $[t^j]P(t)$ are included in `data/witness_generating_function.json`.

The denominator has a pole of exact order 17 at $t = 1$, since $P(1) = \delta_{16} = 16!/51840 > 0$. Equivalently, the sequence satisfies

$$\sum_{j=0}^{17} (-1)^{17-j} \binom{17}{j} w_{m+j} = 0 \quad (m \geq 0). \quad (42)$$

Thus the original graph-theoretic question yields an explicit integer sequence with a rational generating function, a constant-coefficient recurrence, and a complete polynomial asymptotic expansion. No OEIS accession or novelty claim for this derived sequence is assumed.

9 Verification, reproducibility, and proof boundaries

The proof has three logically different components. The explicit C_{17} arithmetic already disproves the conjecture. The symbolic factorization and its sign proof treat every $n \geq 17$. Finally, a finite set of rational inequalities and integer identities settles the cases $3 \leq n \leq 16$. Only the last component needs finite certificate checking, and those certificates are fully specified in the report and accompanying files.

9.1 What was executed

The standard-library program `verify.py` was run with full cycle rows through $n = 200$. It checks the first-failure classification, all nine positive cone certificates, the formulas for the third iterate, and an independent deletion-contraction generation of the coefficient rows. It also computes the necessary six-entry prefix at $n = 1000, 10000, 10^6, 10^{12}$ and compares the exact third-iterate result with (25). These larger checks are corroboration, not the proof of the universal assertion.

A separate SymPy program, `derive_symbolic.py`, was executed to audit all displayed low-index factorizations, both positive-shift identities, the Pascal neighbor ratios, the asymptotic coefficients, and the rational generating-function data. The recorded environment was Python 3.13.5 and SymPy 1.14.0. The principal checker needs only the Python standard library; SymPy is optional. No Lean formalization or independent human referee verification is claimed.

9.2 Exact arithmetic and computational cost

A whole-row iteration uses $O(n)$ integer operations and $O(n)$ integer storage if only the current row is retained. If entries initially have at most B bits, the largest bit length after j iterations is $O(2^j(B+1))$, because a new value is a difference of products or squares. For a fixed depth, full-row tests are consequently practical at modest n , but integer sizes grow rapidly with depth.

The local witness requires only six initial entries, three shrinking prefix updates, and a constant number of integer operations after those entries are formed. Their bit lengths are $O(\log n)$, and the witness has $O(\log n)$ bits as well, since its degree is fixed. This explains why the $n = 10^{12}$ prefix test is easy without ever constructing a row of length 10^{12} .

Floating-point ratio iteration is unnecessary and can be misleading near cancellation. The programs therefore neither convert large coefficients to machine floats nor decide a sign from a decimal approximation. They also never apply absolute values to an iterate: once a negative entry appears, the infinite-log-concavity property has already failed.

9.3 Rebuilding the package

From the archive directory, run

```
python verify.py
python minimal_verifier.py
python derive_symbolic.py # optional: needs SymPy
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The `-write-data` option regenerates the JSON and CSV artifacts. Sequence values in the JSON certificate are decimal strings, deliberately avoiding precision loss in software that parses JSON numbers as binary floating-point values. The CSV data are exact decimal integers.

The archive contains the article source and PDF, the three verification programs, execution transcripts, a README, a build script, and the coefficient and witness datasets. It contains no third-party font files or copied research-paper PDFs.

10 Consequences and remaining directions

The conjecture fails within a particularly restrictive class: simple, connected, 2-regular graphs. The failure is therefore not a consequence of loops, multiple edges, dense graphs, or complicated graph constructions. On the other hand, it is genuinely a higher-iteration phenomenon: every cycle passes the first two iterations.

Adding s isolated vertices to a graph multiplies its chromatic polynomial by q^s . It only shifts the coefficient sequence, and the shift invariance of $\mathcal{L}$ preserves all failure depths. This supplies further counterexamples, although the cycles already give a connected infinite family.

No analogous inheritance from an arbitrary subgraph is claimed: the presence of a long cycle inside a larger graph does not by itself transfer this coefficient calculation.

The positive cases illustrate the limits of the real-rooted sufficient criterion. By (6),

$$A_n(x) = (1+x)((1+x)^{n-1} - 1).$$

For $n \geq 4$ this has nonreal roots obtained from nonreal $(n-1)$ st roots of unity by subtracting 1. Nevertheless $C_4, \dots, C_{11}$ are infinitely log-concave by their cone certificates. Real-rootedness is therefore sufficient but not necessary in this family.

Several questions are separate from what has been proved. The smallest counterexample among *all* simple graphs could be smaller than C_{12} . Whether every chromatic coefficient sequence is 2-log-concave is not decided by the cycle calculations. One may also ask for a useful structural description of those graphs whose coefficient sequences are infinitely log-concave. These are directions for further investigation, not claims of newly verified open-problem status.

The central methodological lesson is concrete: normalize a tractable family, inspect a few boundary coefficients under the nonlinear operator, and use exact arithmetic to expose a sign change. Here that approach leads beyond a single counterexample to an exact classification and a stable asymptotic obstruction.

A The fourth iterate for the twelve-cycle

For $a = a^{(12)}$, the vector $e = \mathcal{L}^4 a$ has the entries in Table 4. All are nonnegative, and every entry of indices 1 through 12 is positive. Together with Proposition 4.4, this certifies that the negative fifth-iterate entry in Section 6 really is the first failure.

k	$e_k = (\mathcal{L}^4 a^{(12)})_k$
0	0
1	45949729863572161
2	1505318939996586804452286
3	54746933663864342589859620161140
4	15531438327857548927780352470984421505
5	11178866851263718584359046258900106285056
6	102599969549662636115648080319674359475968
7	11246650538385145317068550658400941510656
8	13507131422281444862413759705797857505
9	131200081206735639046019512080640
10	5555828662478863428996336
11	299028575209728
12	1

Table 4: A complete exact certificate that the fourth iterate for C_{12} has no negative entries.

B A short checker for the finite certificates

The following complete Python program computes the finite cases directly from binomial coefficients. It requires only the standard library and prints the exact minimizing ratios or negative entries. It uses no precomputed iterate vectors. The larger `verify.py` adds explicit expected values, data generation, redundant identity tests, and checks that remain enabled under Python's optimized mode.

```

"""Small exact checker for the finite part of the classification."""
from fractions import Fraction
from math import comb

def L(a):
    b = [0] + a + [0]
    return [b[k+1]**2 - b[k]*b[k+2] for k in range(len(a))]

def cone(a):
    return min(a) >= 0 and all(
        a[k]**2 >= 3*a[k-1]*a[k+1] for k in range(1, len(a)-1))

for n in range(3, 18):
    a = [0, n-1] + [comb(n, k) for k in range(2, n+1)]
    for j in range(6):
        bad = [(k, x) for k, x in enumerate(a) if x < 0]
        if bad:
            expected = 5 if n == 12 else 4 if n <= 16 else 3
            assert n >= 12 and j == expected and bad[0][0] == 2
            print(n, "failure", j, bad)
            break
        if cone(a):
            assert 3 <= n <= 11
            ratios = [(Fraction(a[k]**2, a[k-1]*a[k+1]), k)
                      for k in range(1, n) if a[k-1]*a[k+1]]
            print(n, "invariant cone", j, "minimum", min(ratios))
            break
        a = L(a)
    else:
        raise AssertionError((n, "no conclusion"))

```

This finite program is not being asked to establish a statement by checking infinitely many iterates. Its positive exit condition invokes Lemma 5.1; its negative exit condition provides an explicit counterexample. That distinction is the reason the finite certificates are logically sufficient.

References

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